

Steven Schaefer

stschaef@umich.edu · stevenschaefer.net

Education	University of Michigan PhD Candidate in Computer Science and Engineering	Sep 2022 – Present
	Indiana University Doctoral Credits in Mathematics, GPA: 4.0	Sep 2020 – Aug 2021
	University of Michigan B.S. in Honors Mathematics with a Minor in Computer Science	Sep 2016 – May 2020
Publications	Intrinsic Verification of Parsers and Formal Grammar Theory in Dependent Lambek Calculus Steven Schaefer, Nathan Varner, Pedro H. Azevedo de Amorim, Max S. New <i>Proceedings of the ACM on Programming Languages</i>	PLDI 2025
	Logical Relations on Display Steven Schaefer, Johnson He, Max S. New	In preparation
Research	Formal Grammars as Types University of Michigan, Ann Arbor, MI	Jan 2024 – Present
	- Designed a domain-specific type theory for intrinsically-verified parsers (PLDI 2025). - Based on a dependently-typed non-commutative, linear-non-linear type theory. - Implemented in Cubical Agda, with an ongoing port to Lean.	
	Mechanized Category Theory University of Michigan, Ann Arbor, MI	Mar 2023 – Present
Work Experience	- Formalizing displayed category theory and logical relations in Cubical Agda. - Contributions accepted in the Cubical Agda standard repository.	
	Graduate Student Instructor University of Michigan, Ann Arbor, MI	Sep 2023 – Dec 2023
	Taught EECS 483 (Compiler Construction) to 100 students.	
	Assistant Instructor Indiana University, Bloomington, IN	Sep 2020 – June 2021
	Led two recitation sections of M211 (introductory calculus) for 60 students.	
	Hacking for Defense Sponsored by NSA, Indiana University, Bloomington, IN	Jan 2021 – June 2021
	Measured available entropy in the Linux pseudorandom number generator using SystemTap; delivered analysis and findings to the NSA.	
	Math/Computer Science Tutor Self Employed	Feb 2016 – 2022
	Six years of private tutoring across calculus, linear algebra, discrete math, and introductory CS.	
	Software Engineering Intern J.P. Morgan Chase & Co., Ann Arbor, MI & Chicago, IL	Jan 2019 – Dec 2019
Talks	Built and deployed a full-stack NLP/ML application to extract information from unstructured text and auto-populate request forms.	
	Intrinsic Verification of Parsers and Formal Grammar Theory in Dependent Lambek Calculus ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)	Jun 2025
	Intrinsic Verification of Parsers and Formal Grammar Theory in Dependent Lambek Calculus Midwest Programming Languages Summit, University of Chicago	Nov 2024
Service	University Relations Chair Computer Science and Engineering Graduate Student Organization (CSEG)	2025 – Present